

THE UNIVERSITY OF AZAD JAMMU & KASHMIR
MUZAFFARABAD

DATABASE MANAGEMENT SYSTEM LAB

Lab 9 — SQL Joins — Part 2

Self Joins, Multi-Table Joins & Library Assessment

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Course Title	Database Management System (CS-2204)
Submitted To	Engr. Awais Rathore
Lab No.	09
Tool Used	XAMPP (MySQL / MariaDB) with phpMyAdmin

1. Objective

- Implement Self Joins using table aliasing to model parent-child relationships (employee-manager).
- Construct multi-table joins chaining up to four related tables.
- Combine LEFT JOIN with aggregation (SUM, COUNT) and COALESCE to preserve zero-count entities.
- Compare filtering inside the ON clause vs. filtering in the WHERE clause.
- Apply all of the above to a full graded assessment on a Library Management System.

2. Database Structure

This lab reuses the Company database (joins_lab) from Lab 8, then introduces a separate Library Management System (library_lab) for the graded assessment.

2.1 joins_lab (reused from Lab 8)

Table	Purpose
Department, Employee, Project, Assignment	Same company schema as Lab 8

2.2 library_lab (assessment)

Table	Purpose
Author	Author name and country
Book	Book title, genre, price, and linked author
Member	Library member details
Loan	Links members to borrowed books, with loan/return dates

3. Key Tasks / Concepts Implemented

- Task B1-B2: Self join for employee-manager names; employees earning more than their manager.
- Task B3, B8: Cross-department comparisons between employees, managers, and projects.
- Task B4-B5: 3-table and 4-table joins chaining Employee, Assignment, Project, and Department.
- Task B9-B10: Filtering inside ON vs. WHERE; LEFT JOIN + COALESCE for zero-hour employees.
- Assessment Q1-Q10: books with authors, authors with no books, members who never borrowed, active loans, and an emulated FULL OUTER JOIN via UNION.

4. Implementation (phpMyAdmin)

All Part B tasks and the Q1-Q10 assessment were executed and verified using phpMyAdmin under XAMPP.

✓ Showing rows 0 - 9 (10 total, Query took 0.0003 seconds.)

```
SELECT e.EmpName AS Employee, m.EmpName AS Manager FROM Employee e LEFT JOIN Employee m ON e.ManagerID = m.EmpID;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 ▾ Filter rows:

Extra options

Employee	Manager
Ali Khan	NULL
Sara Iqbal	Ali Khan
Hamza Raza	Ali Khan
Ayesha Noor	NULL
Bilal Ahmed	Ayesha Noor
Fatima Sheikh	NULL
Usman Tariq	Fatima Sheikh
Maira Javed	NULL
Zain Abbas	Maira Javed
Nida Yousaf	Maira Javed

Self Join

← Server: 127.0.0.1 » Database: library_lab

Structure SQL Search Query Export Import Operations Privileges Routines Events ▾

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ author	★ Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ book	★ Browse Structure Search Insert Empty Drop	9	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ loan	★ Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ member	★ Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	16.0 KiB	-
4 tables	Sum	27	InnoDB	utf8mb4_general_ci	112.0 KiB	0 B

⬆ ☐ Check all With selected: ▾

Library database structure tab

✓ Showing rows 0 - 7 (8 total, Query took 0.0003 seconds.)

```
SELECT b.BookID, b.Title, a.AuthorName, a.Country FROM Book b INNER JOIN Author a ON b.AuthorID = a.AuthorID;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 ▾ Filter rows:

Extra options

BookID	Title	AuthorName	Country
101	Pride and Prejudice	Jane Austen	UK
102	Emma	Jane Austen	UK
103	Things Fall Apart	Chinua Achebe	Nigeria
104	Norwegian Wood	Haruki Murakami	Japan
105	Kafka on the Shore	Haruki Murakami	Japan
106	Ice-Candy-Man	Bapsi Sidhwa	Pakistan
107	The Reluctant Fundamentalist	Mohsin Hamid	Pakistan
108	Exit West	Mohsin Hamid	Pakistan

✓ Showing rows 0 - 2 (3 total, Query took 0.0004 seconds.)

```
SELECT b.Title, m.MemberName, m.City, l.LoanDate FROM Loan l INNER JOIN Book b ON l.BookID = b.BookID INNER JOIN Member m ON l.MemberID = m.MemberID WHERE l.ReturnDate IS NULL;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 ▾ Filter rows:

Extra options

Title	MemberName	City	LoanDate
Norwegian Wood	Ahmad Raza	Lahore	2024-04-10
The Reluctant Fundamentalist	Sara Imran	Karachi	2024-05-01
Exit West	Fatima Ali	Islamabad	2024-06-01

☐ Show all | Number of rows: 25 ▾ Filter rows:

```
SELECT a.AuthorName, b.Title FROM Author a LEFT JOIN Book b ON a.AuthorID = b.AuthorID UNION SELECT a.AuthorName, b.Title FROM Author a RIGHT JOIN Book b ON a.AuthorID = b.AuthorID;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 ▾ | Filter rows:

Extra options

AuthorName	Title
Jane Austen	Pride and Prejudice
Jane Austen	Emma
Chinua Achebe	Things Fall Apart
Haruki Murakami	Norwegian Wood
Haruki Murakami	Kafka on the Shore
Bapsi Sidhwa	Ice-Candy-Man
Mohsin Hamid	The Reluctant Fundamentalist
Mohsin Hamid	Exit West
Anonymous Writer	NULL
NULL	Mystery Title

3 assessment queries mentioned with screenshots

5. Conclusion

This lab extended join concepts to self-referencing hierarchies and multi-table chains, then proved the skill end-to-end through a full graded Library Management System assessment.